

TASVIRLARNI TAHLIL QILISH VA ANGLASH · Amaliy mashg'ulot (Practice)

11-amaliy mashg'ulot. Harakat, optical flow va kuzatish

2026–2027 o'quv yili

Mavzu. Brightness constancy, optical flow tenglamasi, aperture muammosi; Lucas–Kanade (structure tensor, piramida), Horn–Schunck, Farneback; EPE; background subtraction; α – β filtri, Kalman filter, SORT (Hungarian algorithm).

Maqsadlar. Talaba: optical flow tenglamasi, Lucas–Kanade 2×2 tizimi, Horn–Schunck qadami va Kalman predict/update ni qo'lda bajaradi; OpenCV da siyrak va zich oqimni hisoblab, ma'lum haqiqiy siljish bo'yicha EPE ni o'lchaydi; taxmin buzilishini (katta siljish, yoritish) tahlil qiladi; simulyatsiya qilingan obyektни Kalman filter bilan kuzatadi va izlarni bog'laydi.

Muhit. Python 3.11 yoki 3.12, faqat CPU. `python -m venv .venv`, so'ng `pip install -r requirements.txt` (numpy 2.4.4, scipy 1.17.1, scikit-image 0.26.0, opencv-python-headless 4.13.0.92). Kod boshida `np.random.seed(42)`, `cv2.setRNGSeed(42)`, `cv2.setNumThreads(1)`; har bir tasodifiy jarayon `np.random.default_rng(42)`. Tasvir: `skimage.data.camera()` (512×512 , uint8, internet kerak emas). Sonlar 4 xonagacha.

Baholash (100 ball). A (1–13): har biri 2 ball; B (14–17): har biri 4 ball; C (18–30): 21, 25 va 27-topshiriq 6 balldan, qolganlari 4 balldan. Kod — bitta `solutions.py`, har topshiriq alohida funktsiya.

A. Qo'lda hisoblash

- Brightness constancy.** Pikselda $I_x = 2$, $I_y = -1$, $I_t = 3$. Optical flow tenglamasini yozing; normal flow vektori $d_\perp$ va uning uzunligini toping. Qaysi komponent aniqlanmaydi?
- Hosilalar.** $I_1 = \begin{pmatrix} 10 & 20 & 30 \\ 20 & 30 & 40 \\ 30 & 40 & 50 \end{pmatrix}$ (qator — y , ustun — x), $I_2 = I_1 - 10$. Markazda I_x, I_y (markaziy ayirma) va I_t ; tenglamani yozing. Haqiqiy siljish $(1, 0)$ uni qanoatlantiradimi? Yana qaysi siljishlar?
- Lucas–Kanade.** Oynada (I_x, I_y, I_t) : $(2, 1, -4)$, $(1, 2, -5)$, $(1, 0, -1)$, $(0, 1, -2)$. A , b , $d = A^{-1}b$ va A ning xos qiymatlarini toping; har pikselda qoldiqni tekshiring.
- Structure tensor.** (a) $\text{diag}(50, 48)$; (b) $\begin{pmatrix} 40 & 0 \\ 0 & 0,02 \end{pmatrix}$; (c) $\text{diag}(0,1; 0,05)$. Har biri uchun Harris $R = \det A - 0,04(\text{tr } A)^2$, Shi–Tomasi ($\lambda_{\min} > 1$) qarori va turi (burchak/chegara/tekis).
- Horn–Schunck.** $\bar{u} = 0,5$, $\bar{v} = 0$, $I_x = 1$, $I_y = 2$, $I_t = -3$, $\alpha^2 = 4$. Bir iteratsiyadan keyingi (u, v) .
- Piramida.** Bir qavat $r = 3$ px gacha topadi. 40 px siljishda $L = 3$ piramidaning eng yuqori qavatidagi siljish? $L = 3$ va $L = 2$ yetarlimi? 100 px uchun eng kichik L qancha?
- EPE.** Haqiqiy oqim hamma joyda $(10; 0)$. Baholar: $(10,2; 0,1)$, $(9,9; -0,2)$, $(10; 0)$, $(4; 1)$, $(10,3; 0,4)$. Har EPE, AEPE, mediana va FI ($\text{EPE} > 3$ va $> 5\%$) ulushi.
- Kalman filter.** $F = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $H = (1 \ 0)$, $Q = I$, $R = 16$, $x = (20; 2)^T$, $P = \text{diag}(9; 4)$, $z = 25$. x^- , P^- , S , K , yangilangan x va P .
- α – β filtri. $\alpha = 0,5$, $\beta = \alpha^2/(2 - \alpha)$, $x_0 = 0$, $v_0 = 1$, o'lchovlar $z_1 = 1,4$, $z_2 = 1,8$. Ikki qadam: x_1, v_1, x_2, v_2 .
- Fon modeli.** $B_{t-1} = 80$, $\rho = 0,02$, $\tau = 25$, $I_t = 200$: oldingi planmi, $B_t = ?$ To'xtagan obyekt necha kadrda fonga 90% "singadi" ($1 - (1 - \rho)^n \geq 0,9$)?
- Harakat maydoni.** Kamera oldinga $T_z = 0,4$ m/kadr, $f = 600$ px. $(x, y) = (150; -60)$ px nuqta, $Z = 12$ m: (u, v) ? FOE qayerda? Obyekt tasviri $50 \rightarrow 52$ px (30 kadr/s): to'qnashuvigacha vaqt?
- Bog'lash.** Izlar (qator) va boxlar (ustun) IoU: $\begin{pmatrix} 0,9 & 0,8 & 0 \\ 0,85 & 0,1 & 0 \\ 0 & 0,3 & 0,6 \end{pmatrix}$. Greedy va Hungarian algorithm ($c = 1 - \text{IoU}$) mosliklari, IoU yig'indisi; $\text{IoU} < 0,3$ juftlar rad etiladi.
- Block matching.** 1280×720 kadr, 16×16 blok, qidiruv ± 8 px, har nomzodda 256 ayirma. Bir kadrda va 25 kadr/s da sekundiga amallar soni?

B. Keltirib chiqarish va tushunchalar

- Taylor yoyilmasidan $I_x u + I_y v + I_t = 0$ ni chiqaring; normal flow formulasini isbotlang. $I_2 = 0,8 I_1$ (siljishsiz) bo'lsa, $\|\nabla I\| = 20$, $I_1 = 100$ pikselda qanday "soxta" oqim paydo bo'ladi?
- Lucas–Kanade normal tenglamalarini chiqaring; $A \succeq 0$ ekanini va barcha gradientlar parallel bo'lsa rank $A = 1$ ekanini isbotlang. Nega bu Shi–Tomasi mezoni bilan bir xil?
- Horn–Schunck: Euler–Lagrange tenglamalaridan $\Delta u \approx \kappa(\bar{u} - u)$ yaqinlashuvi bilan iteratsion formulani (Kramer usuli) keltirib chiqaring. $\alpha \rightarrow 0$ da bir qadam nimani beradi?

17. Skalyar Kalman ($F = H = 1, Q = 0$): $1/P_k = 1/P_{k-1} + 1/R$ ni isbotlang; $P_0 \rightarrow \infty$ da baho o'lchovlar o'rtachasi ekanini ko'rsating. Qisqacha: SORT qadamlari; DeepSORT farqi; nega mediana EPE va forward-backward check kerak?

C. Dasturlash

Umumiy yordamchilar (barcha C topshiriqlarida aynan shular):

```
def shift(img, dx, dy): # mazmun (dx, dy) ga siljiydi: haqiqiy oqim = (dx, dy)
    M = np.float32([[1, 0, dx], [0, 1, dy]]); h, w = img.shape[:2]
    return cv2.warpAffine(img, M, (w, h), flags=cv2.INTER_LINEAR, borderMode=cv2.BORDER_REFLECT)

def make_texture(n=256, sigma=3.0):
    t = cv2.GaussianBlur(np.random.default_rng(42).uniform(0, 255, (n, n)).astype(np.float32), (0, 0), sigma)
    return cv2.normalize(t, None, 0, 255, cv2.NORM_MINMAX)

def good_points(img, n=100, border=40): # Shi-Tomasi, chetdan border px ichkarida
    p = cv2.goodFeaturesToTrack(img, n, 0.01, 8); h, w = img.shape
    k = (p[:,0,0] > border) & (p[:,0,0] < w-border) & (p[:,0,1] > border) & (p[:,0,1] < h-border)
    return p[k]

LK = dict(winSize=(21, 21), maxLevel=3,
          criteria=(cv2.TERM_CRITERIA_EPS | cv2.TERM_CRITERIA_COUNT, 30, 0.01))
# hosilalar: cv2.Sobel(I, cv2.CV_64F, 1, 0, ksize=3) / 8 (va 0, 1 uchun)
```

18. **Chiziqlashtirish xatosi.** To'qimani $d \in \{1, 2, 4, 8\}$ px gorizontal siljiting; I_x ni $(I_1 + I_2)/2$ dan, $I_t = I_2 - I_1$. Chetdan 20 px ichkarida median $|I_x d + I_t| / \text{median}|I_t|$ ni chop eting. A2-misolni tekshiring. `shift(tex, 1.3, 0)` va `shift(tex, 1.3125, 0)` farqining maksimumini toping — izohlang.
19. **O'z Lucas-Kanade.** `lk_system(g)` ($g: N \times 3$) — A3-misolni yeching. `lk_point(I1, I2, x, y, win=21)`: iterativ (ko'pi bilan 20, $\|\Delta d\| < 10^{-3}$), oynani `scipy.ndimage.map_coordinates(order=1)` bilan oling. Haqiqiy siljish (1,25; -0,75); nuqtalar `tex.astype(np.uint8)` da, $n = 20$ (chet filtridan keyin qolgani). Median/max EPE va median iteratsiya; `cv2.calcOpticalFlowPyrLK` bilan solishtiring.
20. **Aperture.** 100×100 tasvir: (a) vertikal chegara `img[:,50:]=200`; (b) burchak `img[50:,50:]=200`; (c) tekis; `GaussianBlur(sigma=2)`. (50,50) markazli 21×21 oynada A xos qiymatlari, $\lambda_{\min}/\lambda_{\max}$; `shift(3,2)` dan keyin `calcOpticalFlowPyrLK(maxLevel=0)` ning status va d si.
21. **KLT (amaliy).** `camera()` ni (4,0; -2,5) ga siljiting; 100 nuqta; LK sozlamasi. Nuqtalar soni, kuzatilgani, median, o'rtacha va max EPE.
22. **Piramida.** Xuddi shu nuqtalar, $d \in \{2, 6, 12, 20, 40\}$ gorizontal: `maxLevel=0` va 3 uchun median EPE va status yig'indisi (`winSize=(21,21)`, standart criteria).
23. **Farneback.** `calcOpticalFlowFarneback(I1, I2, None, 0.5, 3, 21, 3, 5, 1.2, 0)`, $d \in \{(1, 0), (3, 2), (8, 0)\}$. Chetdan 20 px ichkarida AEPE, mediana; to'qimali ($|\nabla I| > 5$) va tekis piksellarda AEPE alohida; to'qimali ulush.
24. **Horn-Schunck.** Funksiya `horn_schunck(I1, I2, alpha2, n_iter)`; $\bar{u}$ — yadro $[[1, 2, 1], [2, 0, 2], [1, 2, 1]]/12$ bilan `cv2.filter2D`. A5-misolni tekshiring. Ma'lumot (normallashsiz, past kontrast):

```
rng = np.random.default_rng(42)
I1 = cv2.GaussianBlur(rng.uniform(0, 255, (80, 80)), (0, 0), 3); I2 = np.roll(I1, 1, axis=1)
```

(α^2, n) $\in \{(1, 200), (100, 200), (100, 2000)\}$ uchun $[10:70]^2$ sohada median u va v ; I_1 oralig'ini ham chop eting.

25. **Yoritish va forward-backward.** `camera()`, 200 nuqta (chet filtrsiz), siljish (2,1); ikkinchi kadr (a) o'zgarmagan, (b) $\times 0,8$ (clip, uint8). `status` soni, median/o'rtacha EPE; orqaga LK bilan $\|p_2 - p_0\| < 1$ (ikkala `status = 1`) filtridan keyin nuqtalar soni va o'rtacha EPE.
26. **Global harakat.** `camera()` ni markaz atrofida 2° (`getRotationMatrix2D`) burib, (5, -3) ga siljiting. Mustaqil obyekt: `obj=I1[200:280,200:280]` birinchi kadrda [60:140,360:440] ga, ikkinchisida [80:160,330:410] ga qo'yilsin. 300 nuqta (chet 20), LK, `estimateAffinePartial2D(RANSAC, 1.0)`: inlierlar, burchak, masshtab, markazga nisbatan siljish, obyekt sohasidagi ($x \in [355, 445], y \in [55, 145]$) outlierlar.
27. **Kalman kuzatish.** A8-misolni tekshiring. Obyekt: $(10 + 1,5t; 200 - 0,8t)$, $t = 0..99$; o'lchov shovqini $\sigma = 3$ (`default_rng(42).normal`, shakl (100,2)); 50..59-kadrlarda detektsiya yo'q (faqat predict). Holat $[x, y, v_x, v_y]$, $P_0 = 100I$, $Q = 0,01I$, $R = 9I$, $x_0 = (z_0; 0; 0)$. RMSE: xom va Kalman (ko'rinadigan kadrlar, $t \geq 10$); to'silish paytidagi max/o'rtacha xato; oxirgi tezlik bahosi.
28. α - β filtr. A9-misol. Haqiqat $1,5t$, $t = 0..99$, z = haqiqat $+ \mathcal{N}(0, 3^2)$ (seed 42), $x_0 = z_0$, $v_0 = 0$. $\alpha \in \{0,05; 0,1; 0,2; 0,3; 0,5; 0,8; 1\}$ uchun RMSE; eng yaxshi α va yutuq.
29. **Background subtraction.** Fon: `resize(camera(), (160,120), INTER_AREA)`; 60 kadr, har kadrda $\mathcal{N}(0, 3^2)$ shovqin (bitta `default_rng(42)` har tajriba boshida), obyekt [50:66, x0:x0+16], $x_0 = 10 + 2f$, qiymati 255-fon. `createBackgroundSubtractorMOG2(history, 16, detectShadows=False)`, so'ng 3×3 ochish. $f = 40..59$ bo'yicha o'rtacha IoU, precision, recall; history = 50 va 200. A10-misolni tekshiring.

30. SORT-lite. A12-misolni `scipy.optimize.linear_sum_assignment` bilan tekshiring. Uch obyekt ($t = 0..59$): $(20 + 2t; 60 + 0,5t)$, $(140 - 2t; 60 + 0,4t)$, $(80; 20 + 1,5t)$; detektsiyalar $+ \mathcal{N}(0; 1,5^2)$ (shakl $(60, 3, 2)$), soʻng har kadr tartibi `rng.permutation(3)` bilan aralashtiriladi. Xarajat — Evklid masofa, > 15 rad. Ikki rejim: oldingi kadrda joy va Kalman bashorati ($R = 2,25I$, 27-topshiriq modeli). ID almashishlar soni va oxirgi moslik.

Hujjatlar.

- https://docs.opencv.org/4.x/dc/d6b/group__video__track.html
- https://docs.opencv.org/4.x/d4/dee/tutorial_optical_flow.html
- https://docs.opencv.org/4.x/d7/df3/classcv_1_1BackgroundSubtractorMOG2.html
- https://docs.opencv.org/4.x/d9/d0c/group__calib3d.html
- https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.linear_sum_assignment.html
- https://docs.scipy.org/doc/scipy/reference/generated/scipy.ndimage.map_coordinates.html
- <https://scikit-image.org/docs/stable/api/skimage.data.html#skimage.data.camera>